

5th and 6th Week Summary (10/8/26)

- **Confidence intervals (Estimation)**

An interval calculated from the data, usually of the form: estimate $\pm$ margin of error.

confidence interval for μ is $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$ if

data is coming from normal population having unknown mean μ and known σ . or

large random sample from a non-normal population having unknown mean μ and known σ .

- The confidence interval will have a specified **margin of error** m when the sample size is: $n = \left(\frac{z_{\alpha/2} \sigma}{m} \right)^2$

- **Hypothesis tests** (a.k.a., tests of significance) for *population parameter*

H_0 is called the *null hypothesis* while H_a is called the *alternative hypothesis*.

Usually the null hypothesis is a statement of *no effect* or *no difference*.

The claim about the population that we are *trying to find evidence* for is the alternative hypothesis.

A **test statistic** calculated from the sample data measures how far the data diverge from the null hypothesis H_0 . Define the test statistic $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

Decision Rule: Given fixed $\alpha = P(\text{Type-I Error})$.

$H_a : \mu > \mu_0$, Reject H_0 if $z > z_\alpha$

$H_a : \mu < \mu_0$, Reject H_0 if $z < -z_\alpha$

$H_a : \mu \neq \mu_0$, Reject H_0 if $|z| > z_{\alpha/2}$

- If we reject H_0 when in fact H_0 is true, this is a **Type I error**.

The significance level α of any fixed-level test is the probability of a Type I error.

- If we fail to reject H_0 when in fact H_a is true, this is a **Type II error**, also called β .

For one sided alternative test ($H_a : \mu > \mu_0$ or $H_a : \mu < \mu_0$): $\beta = P\left(Z \leq z_\alpha - \frac{|\mu_0 - \mu_a|}{\sigma/\sqrt{n}}\right)$

For two sided alternative test ($H_a : \mu \neq \mu_0$): $\beta = P\left(Z \leq z_{\alpha/2} - \frac{|\mu_0 - \mu_a|}{\sigma/\sqrt{n}}\right)$

- The **power of a test against any alternative** is 1 minus the probability of a Type II error for that alternative: power = $1 - \beta$.

- **Power analysis** (Sample size determination):

For one sided alternative test ($H_a : \mu > \mu_0$ or $H_a : \mu < \mu_0$): $n = \sigma^2 \frac{(z_\alpha + z_\beta)^2}{\Delta^2}$

For two sided alternative test ($H_a : \mu \neq \mu_0$): $n = \sigma^2 \frac{(z_{\alpha/2} + z_\beta)^2}{\Delta^2}$

